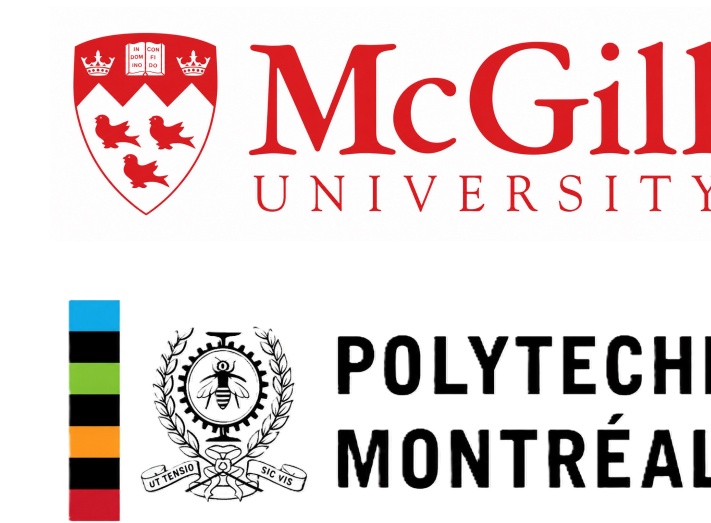


Training Diffusion Language Models for Black-Box Optimization

DiBO: An 8B Diffusion LLM Optimizer for AI-driven Scientific Discovery

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Can an LLM help design a faster-running robot, or discover better DNA sequences to bind a given protein?

Real lab experiments are expensive, and the design[★] → label[★] rule is hidden.

DiBO tackles for AI-driven scientific discovery: learn from limited {design, label} data, then propose better designs.

★ **Design** = the object / configuration to be improved

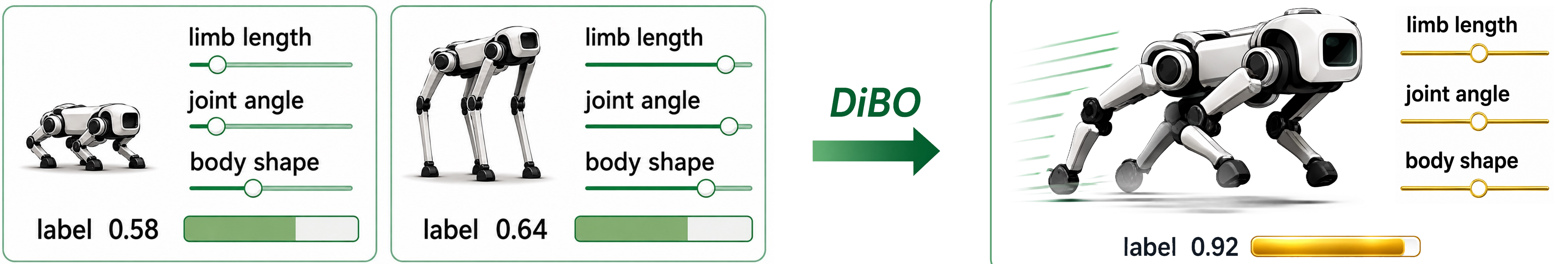
(e.g. Robot Morphology, DNA Sequences, Superconducting Materials, Circuit Layouts)

★ **Label** = how good a design is / how well it performs

(e.g. Robot Speed, Binding Affinity, Superconducting Critical Temperature, Circuit Bandwidth)

Embodied AI

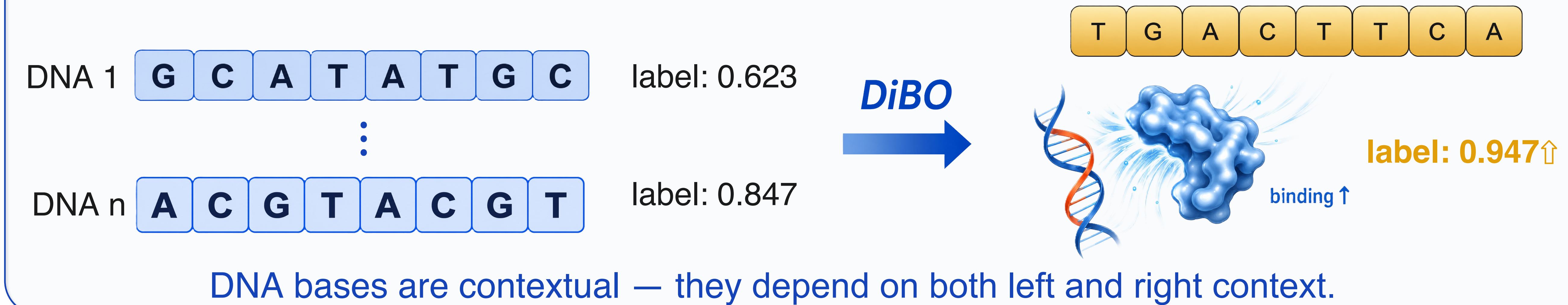
Design Robots to Move Faster and Better



Robot body is a coupled system — each joint affects the whole motion.

Molecular Discovery

Design DNA Sequences to Bind Stronger



DNA bases are contextual — they depend on both left and right context.

Why diffusion LLMs?

Different design components should be modeled **jointly and bidirectionally**, not one by one / purely left-to-right.

Autoregressive (AR) LLM

Sees only from left (unidirectional)

A → C → G → T → ?

Diffusion LLM (dLLM)

Uses left + right (bidirectional) context

[M] C [M] T [M]

1. Data Curation

Unified Prompt-Response Corpus

Notation

q : prompt = task
+ offline examples
+ instruction
 o : response design
 $y(q)$: best label in the prompt
 $y(o)$: label of the response

● Natural Language Prompt (task description, instruction)
● Offline Designs & Associated Labels
● Response Design

Task description

Design a length-8 DNA sequence with high binding affinity.

Offline design-label examples

|design-start| A C G T A C G A |design-end| |label-start| 0.62 |label-end|
|design-start| G C A T A T G C |design-end| |label-start| 0.71 |label-end|

Instruction

Propose a new design with a higher label.

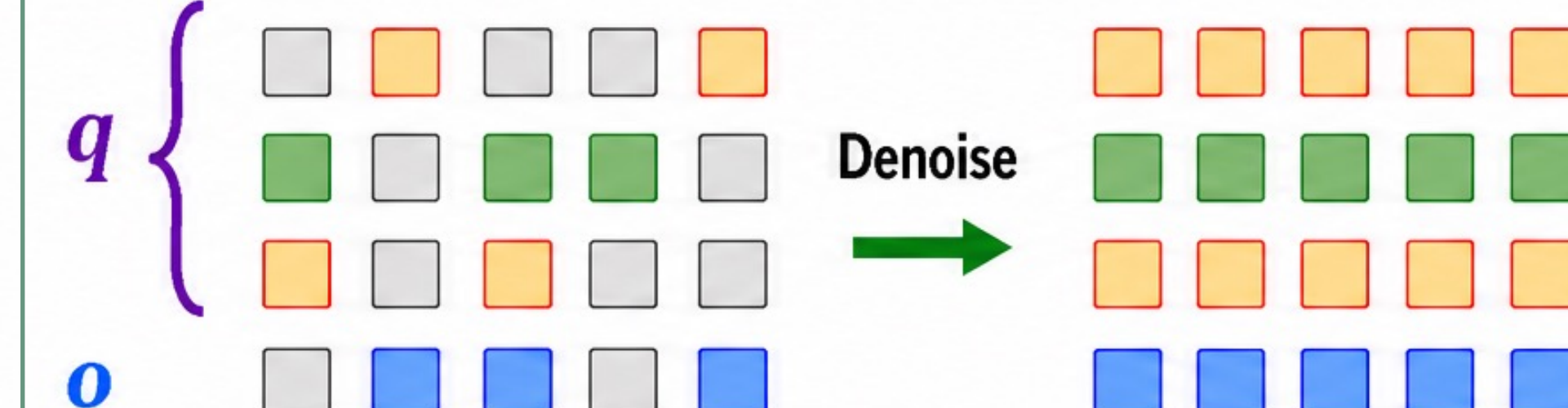
Response design

|design-start| G C G T A C G A |design-end|

How DiBO Works: Data Curation → CPT → SFT → RL

2. CPT

Adapt the LLM to Design-Label Data
Continue pre-training on the unified corpus



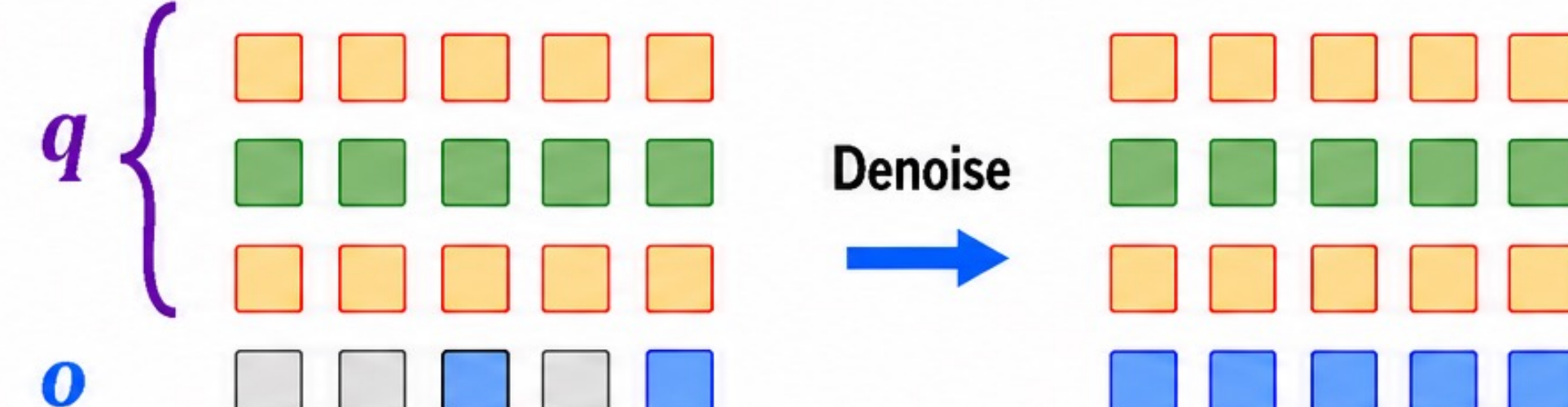
Mask anywhere in q and o .

Tokens of Natural Language Prompt Text Tokens of Offline Designs & Associated Labels Tokens of Response Design Masked Tokens

3. SFT

Imitate High-Label Responses

Train on improved responses: $y(o) > y(q)$.



Keep q visible; randomly mask response tokens in o only.

4. RL

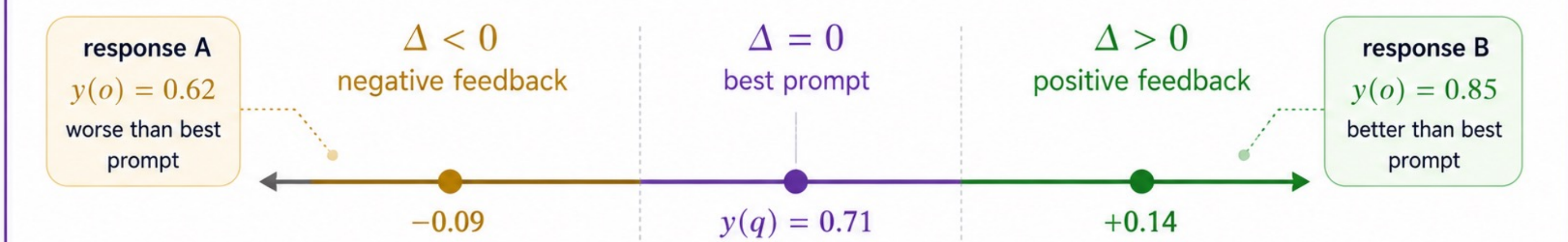
Learn From Both Positive and Negative Feedback



label gap: $\Delta = y(o) - y(q)$
reward: $r(q, o) = \Delta / \sigma$



σ : reward standard deviation



RL uses the **sign** and **magnitude** of the label gap Δ .
Larger $|\Delta|$ → stronger learning signal; $\Delta = 0$ → no update.



SFT imitates only $y(o) > y(q)$; **RL** learns from **both sides** of $y(o) - y(q)$.

SOTA Comparisons

Method	Ant	D'Kitty	TF Bind 8	TF Bind 10	Mean ↑	Rank Mean ↓	Rank Median ↓
$\mathcal{D}(\text{best})$	0.565	0.884	0.439	0.511	—	—	—
Grad-mean	0.709	0.920	0.843	0.736	0.802	4.25	3.5
Grad-EI	0.655	0.923	0.864	0.727	0.792	4.25	4.0
UniSO-T	0.635	0.939	0.836	0.522	0.733	6.75	6.5
ExPT	0.929	0.950	0.810	0.703	0.848	4.0	4.0
DiBO	0.932	0.912	0.946	0.741	0.883	3.5	1.0

Ablation Studies



1. Bidirectional Generation Matters

AR: A → C → G → T → ?
dLLM: [M] C [M] T [M]

+8.0 pts

on TF Bind 8

+7.1 pts

on TF Bind 10

matched dLLM vs AR LLM



2. Delimiters Make Fields Readable

|design-start| |design-end|
|label-start| |label-end|

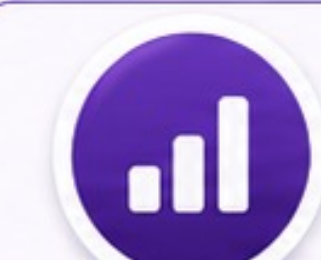
+18.2 pts

on Ant after CPT

+2.4 pts

on TF Bind 8

Structure tokens separate designs from labels.



3. Every Stage Matters

Full CPT + SFT + RL is best

Full: TF8 0.946 · Ant 0.932

Drop stage	TF8 Δ	Ant Δ
w/o CPT	−0.5 pts	−1.9 pts
w/o SFT	−1.8 pts	−7.4 pts
w/o RL	−0.7 pts	−0.3 pts
RL-only	−50.9 pts	invalid



RL-only fails on Ant with invalid output.